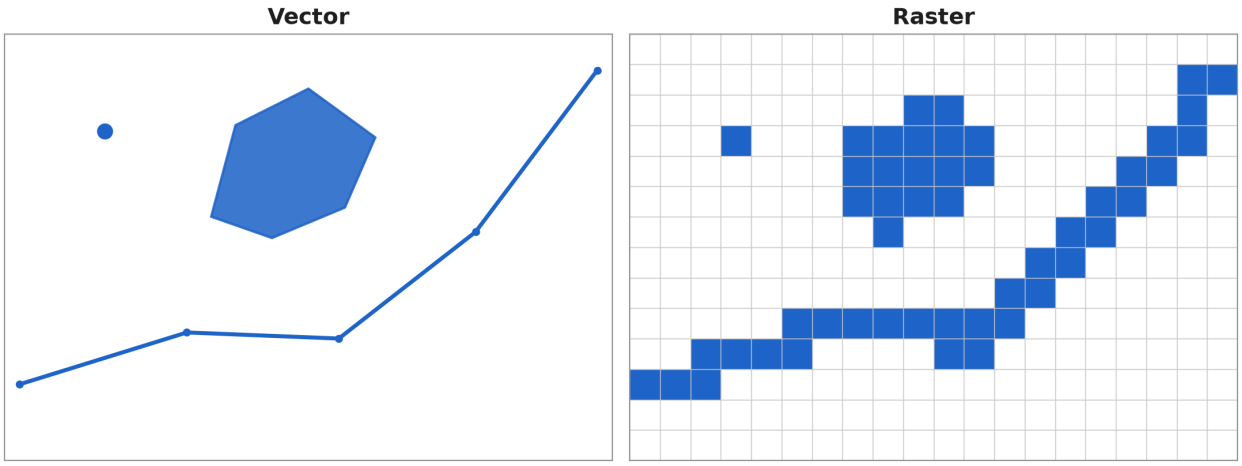
# 2.3 Rasters vs. vectors

A lot of spatial data can be represented as vector data as well as raster data. The choice between the two types depends on the **use case**.

A map with, for example, a boundary stone (point), a road (line) and a lake (polygon) would look like this as vector data (left) and as raster data (right):




| | RASTER | VECTOR |
|---|---|---|
| **Spatial definition** | Based on a matrix of cells | Based on geometric objects |
| **Spatial positioning** | Each raster cell is implicitly given coordinates. This is based on the coordinates of the origin, the spatial resolution (real-world size of each cell) and the position of each cell in the matrix (row and column number). | Coordinates are explicitly stored for each point (or vertex). |
| **File size** | Usually larger files | Usually smaller files |
| **Accuracy** | Resolution-dependent, generalisation at cell level (loss of information) | No technical limits on accuracy |
| **Suitable for** | Continuous and discrete data | Discrete data (continuous data only in a simplified form, e.g. contour lines) |
| **Attribute information** | Raster cells can only contain numeric values. The number of values per cell is limited to one per *image band*. | Each object can have many attribute fields: text, numbers, dates… |
| **Spatial analyses** | Often faster because of the simple data structure | Often slower because of the more complex data structure |